

CFA LEVEL II FIXED INCOME: THE ARBITRAGE-FREE VALUATION BLUEPRINT



LAW OF ONE PRICE

1. THE FOUNDATION OF NO-ARBITRAGE

- **ARBITRAGE**: No initial outlay, riskless profit. Markets adjust to eliminate.
- **LAW OF ONE PRICE**: Identical cash flows = same price. Else, trade until convergence.



- **VALUE ADDITIVITY**: Value of whole = sum of parts (zero-coupon components).

2. ARBITRAGE-FREE SPOT RATE VALUATION

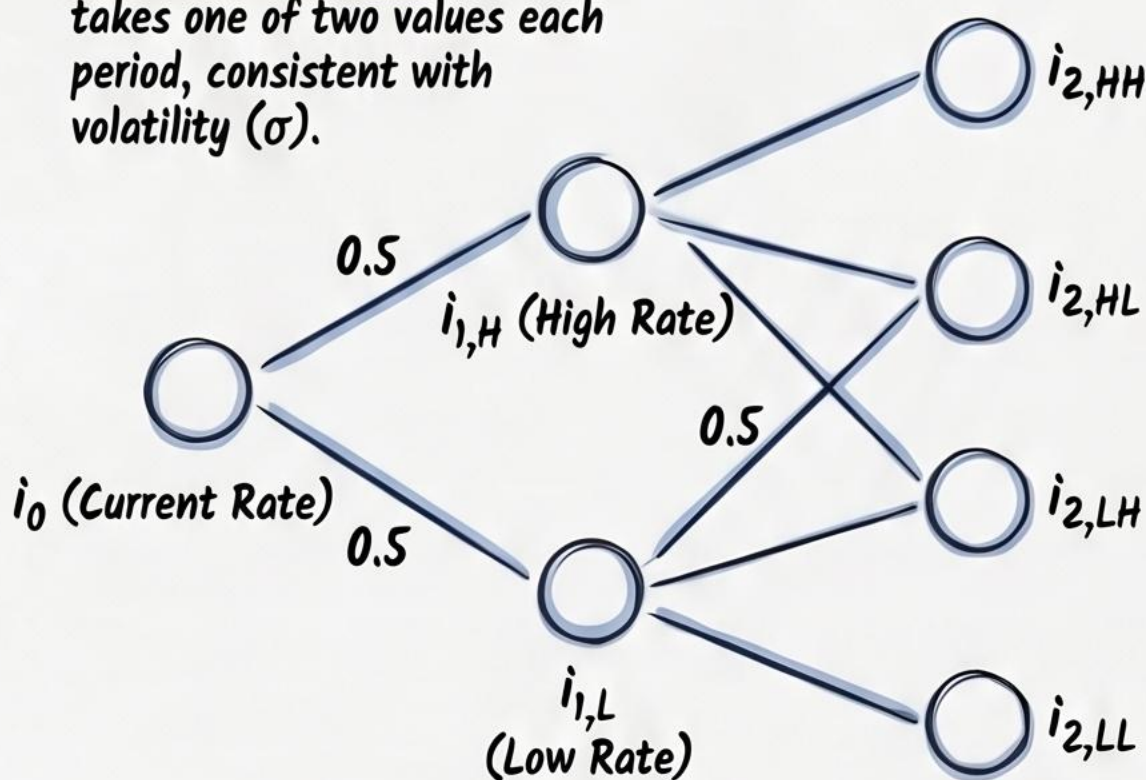
$$PV = \sum_t \left[\frac{CF_t}{(1 + z_t)^t} \right] + \left[\frac{FV}{(1 + z_N)^N} \right]$$



- **SPOT RATES** (z_N): Annualized return on default-risk-free zero-coupon bond. Most accurate time value.
- **BOOTSTRAPPING**: Iterative process to derive spot rates from par curve.

3. THE BINOMIAL INTEREST RATE TREE

- **BINOMIAL LATTICE**: Short rate takes one of two values each period, consistent with volatility (σ).



- **LOGNORMAL RANDOM PROCESS**: Rates follow lognormal walk, ensuring non-negativity, higher volatility at higher rates.

TREE CALIBRATION



- Iterative process to match model prices with observed market prices for benchmark bonds.
- **VOLATILITY IMPACT**: Higher σ spreads tree. If $\sigma = 0$, tree collapses to single path (forward curve).

4. VALUATION METHODOLOGIES

$$Value_{node} = \frac{0.5 \times (V_H + C) + 0.5 \times (V_L + C)}{1 + i}$$

BACKWARD INDUCTION ("WORK BACKWARD")



- **PATHWISE VALUATION**: Calculate PV for every possible path, then average all path values.
- **MONTE CARLO SIMULATION**: For complex securities, simulate many random paths to approximate value.



5. TERM STRUCTURE MODELS & EXAM TRAPS

- **CIR VS. VASICEK**: Both mean-reverting. CIR: Volatility increases with rate level. Vasicek: Constant volatility.



- **YTM TRAP**: Common error to discount all cash flows at single YTM. Must use specific spot rate (z_N).



- **EMBEDDED OPTIONS**: Spot rate valuation fails. Binomial tree required to check for "exercise" at each node.